MICB 306 – Molecular Virology
PRACTICE ASSESSMENT STYLE QUESTIONS

TRUE/FALSE and EXPLAIN QUESTIONS

For the following statements, **indicate if they are true or false**. Using your knowledge of the general principles of molecular virology, provide a **short rationale** for your answer elaborating why you think a statement is true or false, **providing examples where possible**. (2 marks each)

A. Measles virus primarily targets infection of the respiratory epithelium which contributes to the systemic spread of the virus in the host which leads to distant infection of skin keratinocytes and neurons in the brain.

FALSE
Spread of the measles virus within the host occurs because the virus infects local DCs in the respiratory tract. These DC's travel to local lymph nodes, releasing virions into the blood. This is primarily how the virus spreads to infect keratinocytes and other local lymphoid cells in the skin. Also, students may mention that the infected DC's are likely responsible for CNS neuron infection across the blood brain barrier.

Students need to mention DC infection, but need to briefly outline how DC infection leads to spread for the full 1.5 marks in their reasoning.

B. All viruses classified as Class III, IV, V, and VI in the Baltimore classification must encode an RNA dependent RNA polymerase.
FALSE – this is because reverse transcribing viruses can have an RNA genome (Class VI), but do not require RdRP. Instead they have a DNA intermediate to synthesize RNA.

Only viruses with dsRNA genomes (Class II) or (-)/(+) sense ssRNA genomes (Class IV and V) require RdRP to synthesize RNA to replicate their genomes and to synthesize vmRNA that is translation ready. Importantly, they must encode this enzyme because **no host cellular enzymes use an RNA template to synthesize complementary RNA.** The rationale ought to include that the host doesn't encode RdRP

C. Consider an *in vitro* model for influenza A infection. Pre-treating human respiratory epithelial cells grown in culture with an enzyme that cleaves carbohydrates would prevent infection with Influenza A.

TRUE
Removing carbohydrates would remove the sialic acids from the surface of the host cell. Sialic acids are required for Influenza virion attachment via HA.

D. Airborne viruses transmitted via the respiratory route as aerosolized particles are very contagious, and upon infection, can have an increased potential to cause systemic spread within the host compared to airborne viruses transmitted in droplets.

TRUE
Aerosolized particles remain airborne for longer and are smaller. Because they are smaller, one can predict that they are more likely to bypass getting trapped in mucous, can settle deeper into the lungs, and can have an increased potential to access and infect the alveolar epithelium. This site of infection in the lower respiratory tract could more easily lead to systemic spread through the blood because the capillary bed is in such close proximity to the alveolar epithelium. Or, one can say this epithelial layer is so thin and is easily damaged due to local inflammation, which can lead to an increased chance of hematogenous spread.

'VIRAL SLEUTHING' SHORT ANSWER QUESTIONS.

A. A newly discovered virus is found to have a very small, ssDNA genome encoding only a few viral proteins. Sequencing and experimental data demonstrate that the viral proteins encoded in the genome include two capsid structural proteins, a non-structural protein that functions as a scaffold during capsid assembly, and one enzyme that functions exclusively as a protease. Based on this information, where in the host cell would you predict this virus replicates it's genome? Use the information provided in the question to explain your reasoning. (3 marks)

Location – nucleus. (1 mark)
Rationale: (2 marks)
If this virus were to replicate in the cytoplasm **it would need to encode more proteins/enzymes required for genome replication and transcription of the genome (ex: A DNA dependent polymerase)**. It specifies in the question that there is only one enzyme and it functions as a protease. Thus, this virus must rely on nuclear entry **and use of the host replication machinery**.

Note: It is not enough to assume that because it is a DNA virus that it will go to the nucleus. If students only say this as a rationale, they should only be awarded 1 mark for the rationale. There is more pertinent information provided to guide the prediction, students should incorporate information about what is encoded in the genome in their reasoning to receive full marks.

B. Another new pathogenic virus enters a human host via the respiratory tract, has a non-segmented, (+) sense, ssRNA viral genome, a helical nucleocapsid and two spike proteins that are embedded in its viral envelope. Case reports have documented that infections can cause difficulty breathing, swollen lymph nodes, and severe stomach cramps. Make predictions and explain your rationale whether this virus…

a. …causes **Local** or **systemic** infection? *(Circle or highlight, and explain, 1 mark)*

Highlight/circle 0.5 marks, explanation 0.5 marks
The explanation ought to include that there are reported symptoms in the gut, so it is likely that the infection spread there.
Note: The fact that there is swelling in the lymph node only implies that there is an adaptive immune response; this doesn't indicate that immune cells were themselves infected. I would deduct 0.25 if they equate swollen lymph node with infection of DC's or lymphocytes.

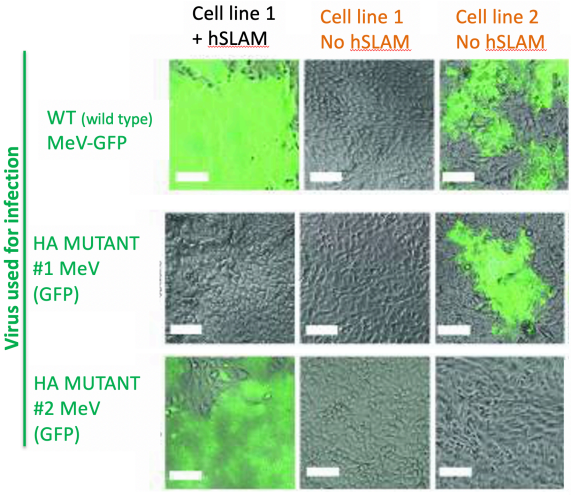
b. …is likely only shed via respiratory secretions? *(Yes or no, and explain, 1 mark)*

No (0.5 marks)
Explanation (0.5 marks) - Since the prediction is that the infection spreads to the gut, it is likely that the virus can also shed via the gastrointestinal tract (feces).

c. …is more or less likely to be easily transmitted via respiratory secretions in drier colder climates. *(Yes or no, and explain, 1 mark)*

Yes/More (0.5 marks)
Explanation (0.5 marks) – in drier climates the respiratory droplets get smaller (water evaporates) and lighter allowing the droplets to spread farther without falling to the ground. Students may also say colder temperatures mean people congregate indoors which increases the chance of human-human transmission. I would accept each of those for the 0.5 mark, and students that mention both are stellar!

DATA ANALYSIS:



Recall the experiment where researchers generated two different mutant measles viruses (MeV) with mutated HAs and engineered the viruses to also express GFP, so if viruses could enter and replicate, the infected cells would glow green (see the data to the left as a reference). They used an *in vitro* infection model to look at how these mutant measles viruses infected different cell types (Cell line 1 and 2) compared to wild type measles virus.

As a follow-up experiment, the researchers tested the two mutant viruses (HA MUTANT #1 and HA MUTANT #2) *in vivo* using a non-human primate animal model. They were able infect three groups of animals infected with each type of virus and observe viral loads and clinical outcomes. Below is a representation of the data they collected. In the last column explain why the researchers obtained this result.

	Animal group 1	Animal group 2	Animal group 3	For each of the measurements (1. viral load in the blood; 2. Rash; 3. viral particles shed), explain these results, and relate the data to how MeV infects a host. (3 marks each)
The researchers measured the following:	Wild-type MeV-GFP	HA Mutant #1 MeV	HA Mutant #2 MeV	
1. Viral particles detected in the blood.	Yes	No	Yes	***Animal Group 1 WT virus*** – WT virus with WT HA can bind SLAM on DC's/lung macs which travel to local LN's and can get into the circulation. ***Animal Group 2 Mutant#1*** – Mut#1 HA CANNOT bind SLAM. Therefore in vivo there will be no infection of DC's/lung macs, so no virus will travel to LN and get in the blood. ***Animal Group 3 Mutant#2*** – Mut#2 HA can still bind SLAM so it is not surprising that virus can get to the blood similar to WT.
2. Skin rash clinical symptoms.	Yes	No	Yes, but only a mild rash. Not as severe as wild-type MeV-GFP	***Animal Group 1 WT virus*** – The WT virus disseminates to the skin, where infection and typical rash symptoms occur. ***Animal Group 2 Mutant#1*** – Mut#1 HA CANNOT bind SLAM, therefore no dissemination and infection/immune reaction in the skin, so no rash. ***Animal Group 3 Mutant#2*** – Mut#2 HA can still bind SLAM, therefore dissemination occurs as normal. Rash is less severe because there is some SLAM-mediated infection, but this mutant can't bind the other receptor – this must decrease some of the infection in the skin *(some students may say directly that this mutant can't bind nectin 4 so can't infect keratinocytes – this is ok too for them to make that conclusion based on what we know now).*
3. Viral particles shed and exited	Yes	No	No	***Animal Group 1 WT virus*** – WT virus has WT HA which can bind both SLAM and the then unidentified other receptor (nectin-4) which is required to infect

the respiratory tract.				respiratory epithelium (stage 3 of MeV infection) so new virions can be produced in the lung and exited! ***Animal Group 2 Mutant#1*** – Mut#1 HA CANNOT bind SLAM, therefore no dissemination, no late stage infection of the respiratory epithelium, no respiratory shedding. ***Animal Group 3 Mutant#2*** – Mut#2 HA can still bind SLAM, so dissemination occurs and infected lymphoid cells travel back to the lung BUT this mutant cannot bind to the respiratory epithelium so no infection, no respiratory shedding!